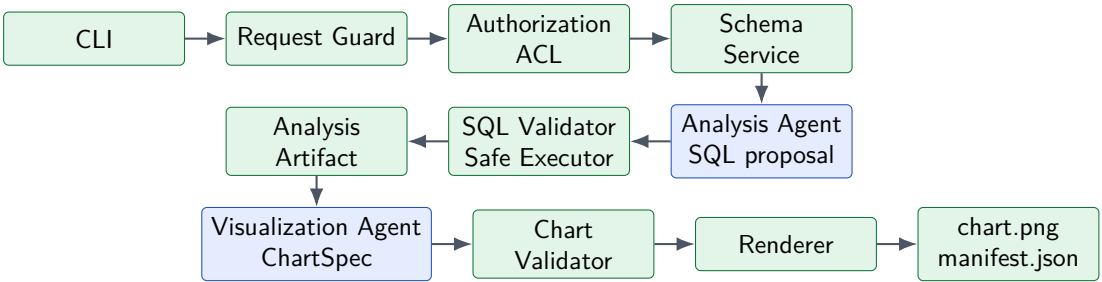
High-Level Architecture



Green boxes are deterministic. Blue boxes are LLM-backed semantic planners.

LLM Agents

Analysis Agent

- Input: user question plus authorized schema.
- Output: structured `AnalysisPlan`.
- May return `query`, `clarification`, or `unsupported`.
- Does not execute SQL.
- SQL is validated and executed by deterministic services.

Visualization Agent

- Input: row-free visualization context.
- Output: declarative `ChartSpec`.
- Does not generate plotting code.
- ChartSpec is validated before rendering.

Design Decisions I

Decision	Reason
Reject unsafe intent early	Request guard rejects obvious destructive or instruction-overriding intent before schema access or LLM calls.
Authorize before database access	ACL policy runs before database connection, schema inspection, SQL generation, and execution.
Validate SQL after LLM proposal	SQL validator allows only read-only SELECT statements over authorized schema objects.
Validate charts before rendering	ChartSpec is validated deterministically before rendering; malformed or misleading specs are rejected.

Design Decisions II

Decision	Reason
Structured outputs, not free text	Agents must return `AnalysisPlan` or `ChartSpec`; deterministic code can validate and reject malformed model output.
Clarification before guessing	When key business meaning is missing, the Analysis Agent asks the user and the CLI reruns with the answer instead of inventing assumptions.
Row-free visualization context	The visualization LLM gets column profiles and aggregate statistics, not raw rows or potentially malicious cell values.
Debuggability without default leakage	`--save-llm-io` records prompts and structured outputs only when explicitly requested, supporting review without always storing prompts.

Future Improvements

- **Authorization module:** replace the demo username-based ACL with real authentication, roles, audit logs, and possibly table/column-level access.
- **Relationship catalog:** persist trusted join hints in config when databases do not declare foreign keys, instead of relying on users to repeat them in every prompt.
- **Large-schema retrieval:** avoid sending every table to the Analysis Agent by first selecting relevant tables/columns deterministically or with a smaller schema-search step.